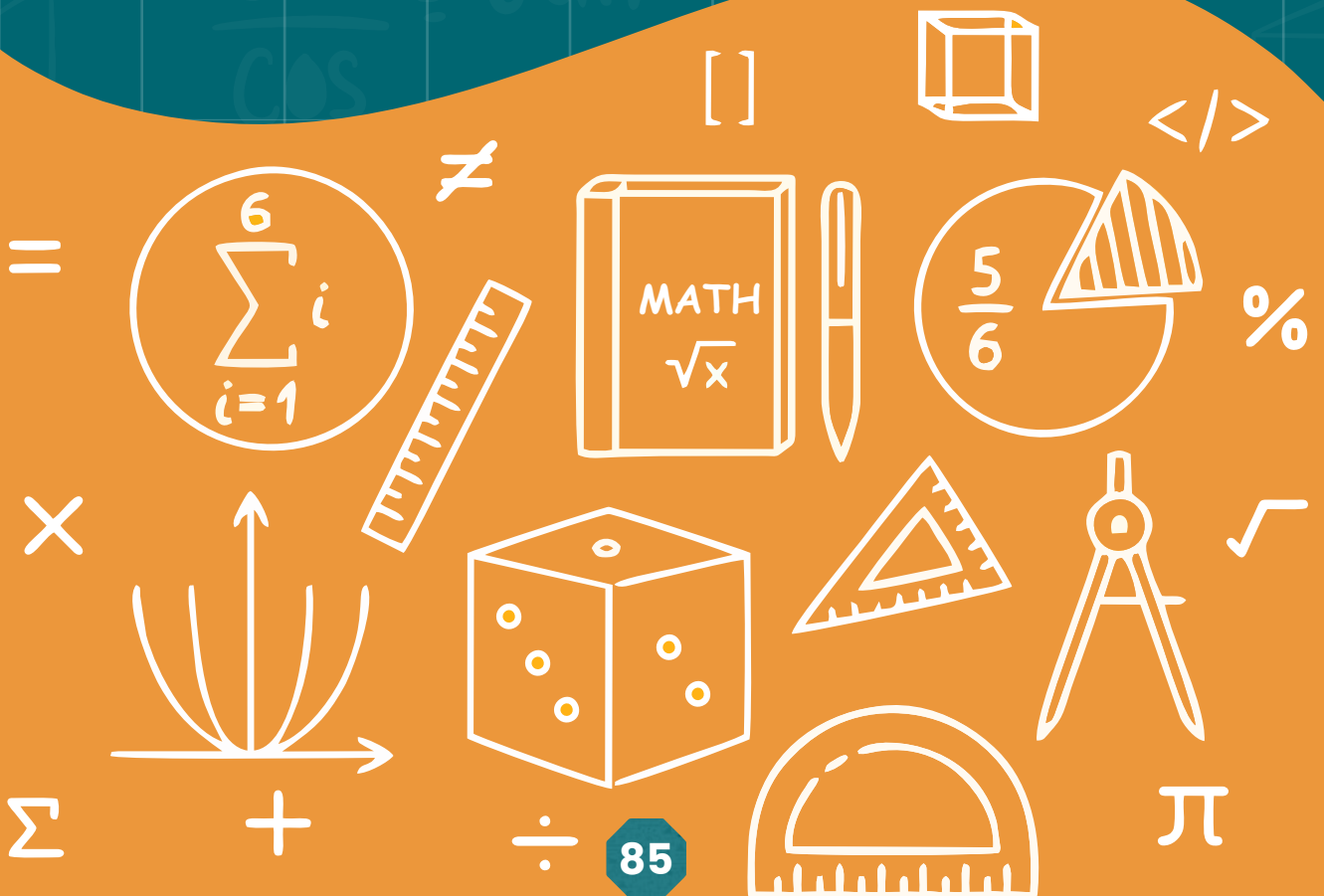


UNIT 8

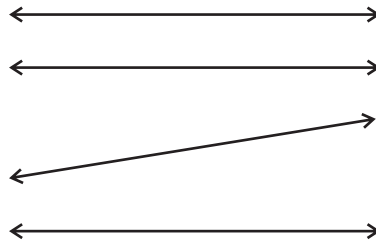
PRACTICAL GEOMETRY



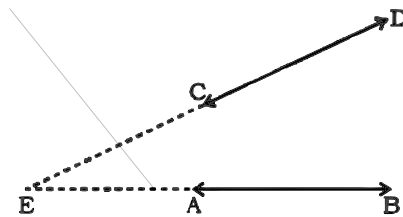
CONSTRUCTION OF QUADRILATERALS

When we draw two lines in a plane, they may be either

- (i) Intersecting lines at a point
- (ii) Parallel lines
- (iii) Neither intersecting nor parallel but in such a state that they will intersect when produced.

**Converging lines:**

Two lines or line segments in a plane which are neither intersecting nor parallel but they will intersect at a point when produced are called converging lines.



To find the measure of an angle between two converging lines without producing them:

Given: Two converging lines $\overleftrightarrow{AB}$ and $\overleftrightarrow{CD}$

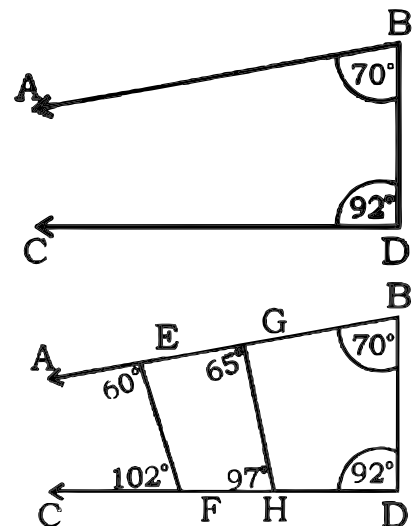
Required: To find the measure of the angle between $\overleftrightarrow{AB}$ and $\overleftrightarrow{CD}$ without producing them.

Steps of construction:

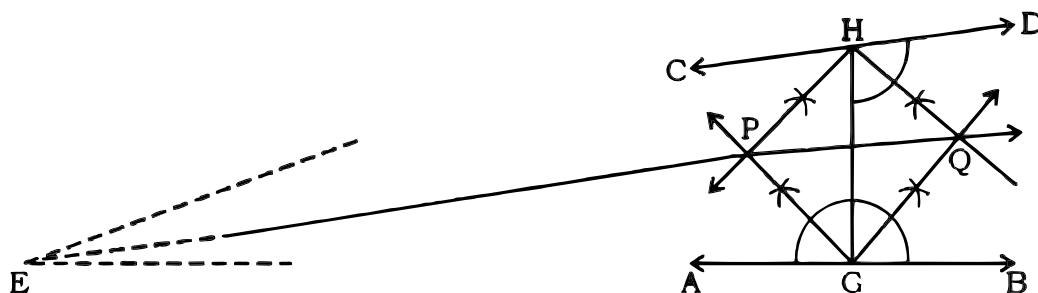
1. Join B to D (Points on the diverging side)
2. Measure $\angle ABD$ with the help of a protractor, we have $m\angle ABD = 70^\circ$
3. Measuring $\angle CDB$ with a protractor, we have $m\angle CDB = 92^\circ$

4. Therefore, the measure of angle (say x) between the converging lines is:

$$\begin{aligned}
 m\angle x &= 180^\circ - (m\angle ABD + m\angle CDB) \\
 &= 180^\circ - (70^\circ + 92^\circ) \\
 &= 180^\circ - 162^\circ \\
 &= 18^\circ
 \end{aligned}$$



Bisect the angle between two converging lines without producing them



Given: Two converging lines $\overleftrightarrow{AB}$ and $\overleftrightarrow{CD}$

Required: To bisect the angle (that would form on producing the converging sides of lines) without producing them.

Steps of construction:

1. At a convenient place, draw a transversal intersecting $\overleftrightarrow{AB}$ at G and $\overleftrightarrow{CD}$ at H.
2. Draw the bisectors of $\angle AGH$ and $\angle CHG$ intersecting each other at point P.
3. Draw the bisectors of $\angle BGH$ and $\angle DHG$ intersecting each other at point Q.
4. Draw $\overline{QP}$ and produce it to any point E.
 $\overleftrightarrow{QE}$ is the required bisector of the angle between two converging lines without producing them.

CONSTRUCT A SQUARE

- (a) Measure of its diagonal is given.
- (b) Difference between the measures of a diagonal and a side is given.
- (c) Sum of the measures of a diagonal and a side are given.

Construct a square when measures of its diagonal is given

Example 1. Draw square ABCD whose diagonal measures 6 cm.

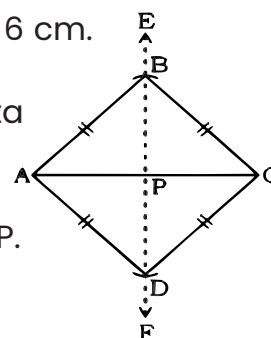
Given: In square ABCD, $m\overline{AC} = 6$ cm

Required: To construct a square ABCD from the given data

Steps of construction:

1. Draw $\overline{AC}$, such that $m\overline{AC} = 6$ cm
2. Draw $\overleftrightarrow{EF}$, the right bisector of $\overline{AC}$ intersecting it at point P.
3. Take P as centre and with radius $m\overline{PA}$ (or $m\overline{PC}$), draw arcs on both sides of $\overleftrightarrow{AC}$ cutting $\overleftrightarrow{EF}$ at B and D.
4. Draw $\overline{AB}$, $\overline{BC}$, $\overline{CD}$, and $\overline{DA}$.

Now ABCD is the required square.



Construct a square when difference between the measures of a diagonal and a side is given

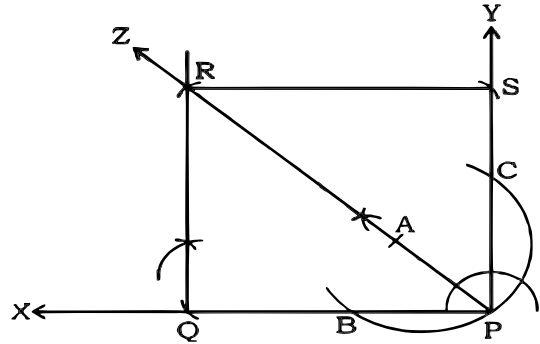
Example 2. Construct square PQRS when the difference between its diagonal and a side is 3 cm.

Given: In square PQRS, difference between its diagonal and one side is 3 cm.

Required: To construct square PQRS from the given data.

Steps of construction:

1. Draw a ray $\overrightarrow{PX}$.
2. At point P draw $\overrightarrow{PY} \perp \overrightarrow{PX}$.
3. Draw $\overrightarrow{PZ}$, the bisector of $\angle XPY$.
4. From $\overrightarrow{PZ}$, cut off $\overline{PA}$ such that $m\overline{PA} = 3$ cm.
5. Taking point A as centre and with radius 3 cm, draw an arc cutting $\overrightarrow{PX}$ at B and $\overrightarrow{PY}$ at point C.
6. Taking points B and C as centres and with the same radius (i.e., 3 cm), draw arcs cutting $\overrightarrow{PX}$ at Q and $\overrightarrow{PY}$ at point S.
7. Taking points Q and S as centres and with radius $m\overline{PO}$ (or $m\overline{PS}$), draw arcs cutting one another and $\overrightarrow{PZ}$ at point R.
8. Draw $\overleftrightarrow{QR}$ and $\overleftrightarrow{SR}$. Now PQRS is the required square.



Construct a square when the sum of measures of its diagonal and side is given.

Example. Construct square ABCD, when the sum of its diagonal and one side is 10 cm.

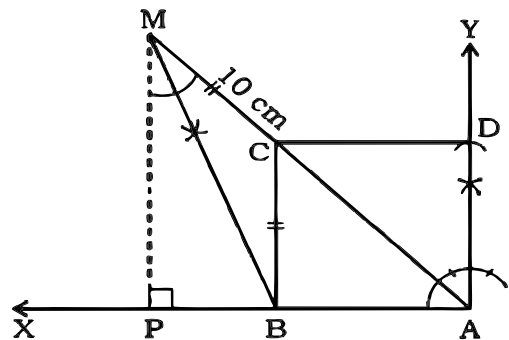
Given: In square ABCD, the sum of its diagonal and a side is 10 cm.

Required: To construct square ABCD from the given data.

Steps of construction:

1. Draw a ray $\overrightarrow{AX}$.
2. Draw $\overrightarrow{AY} \perp \overrightarrow{AX}$.
3. Draw $\overline{AM}$, the bisector of $\angle XAY$ such that $m\overline{AM} = 10$ cm.
4. Draw $\overline{MP} \perp \overrightarrow{AX}$.
5. Draw $\overline{MB}$, the bisector of $\angle AMP$ meeting $\overrightarrow{AP}$ at point B.
6. Through point B, draw $\overline{BC} \parallel \overline{MP}$ meeting $\overline{AM}$ at point C.
7. From $\overline{AY}$, cut off $\overline{AD} \cong \overline{BC}$.
8. Draw $\overline{CD}$.

Now quadrilateral ABCD is the required square



CONSTRUCT A RECTANGLE

- (a) When two sides are given
- (b) When the diagonal and a side given

When two sides are given

Example 1.

Construct rectangle $XYZW$ in which $\overline{mXY} = 6\text{ cm}$ and $\overline{mYZ} = 3.5\text{ cm}$

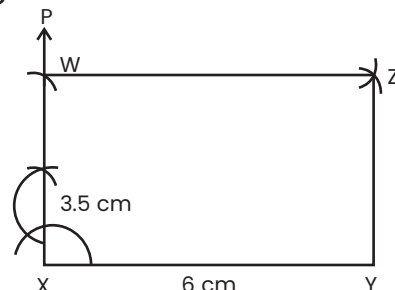
Given: $\overline{mXY} = 6\text{ cm}$, $\overline{mYZ} = 3.5\text{ cm}$, are sides of a rectangle $XYZW$.

Required: To construct rectangle $XYZW$ from the given data.

Steps of construction:

1. Draw $\overline{XY}$, 6 cm long.
2. At point X , draw $\overrightarrow{XP} \perp \overline{XY}$.
3. Take point X as centre and with radius 3.5 cm, draw an arc to cut $\overrightarrow{XP}$ at W .
4. Take point W as centre and with radius $\overline{mXY}$ draw an arc.
5. Take point Y as centre and with radius $\overline{mXW}$ draw an arc which cuts the previous arc at Z .
6. Draw $\overline{WZ}$ and $\overline{YZ}$.

Now $XYZW$ is the required rectangle

**When the diagonal and a side given****Example 2.**

Construct rectangle $ABCD$ in which diagonal $\overline{mAC} = 10\text{ cm}$ and a side $\overline{AB} = 8\text{ cm}$.

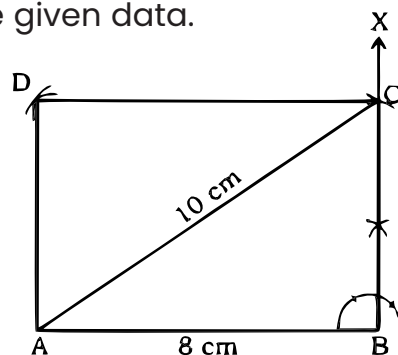
Given: In rectangle $ABCD$, $\overline{mAC} = 10\text{ cm}$ and $\overline{mAB} = 8\text{ cm}$.

Required: To construct a rectangle $ABCD$ from the given data.

Steps of construction

1. Draw $\overline{AB}$, 8 cm long.
2. At point B , draw $\overrightarrow{BX} \perp \overline{AB}$.
3. Take point A as centre, with radius 10 cm, draw an arc which cuts $\overrightarrow{BX}$ at C .
4. Join point A to C (Draw $\overline{AC}$).
5. Take point A as centre, with radius $\overline{mBC}$, draw an arc.
6. Take point C as centre, with radius $\overline{mAB}$, draw another arc which cuts the previous arc at point D .
7. Draw $\overline{AD}$ and $\overline{CD}$.

Now $ABCD$ is the required rectangle.

**CONSTRUCT A RHOMBUS**

(a) When one side and the base angle are given.

(b) When one side and a diagonal are given.

When one side and the base angle are given**Example 1.**

Construct Rhombus $ABCD$ whose sides are 5 cm each and measure of base angle is 60° .

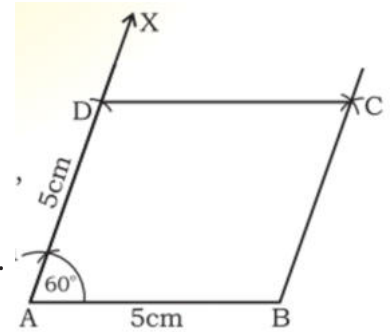
Given: In Rhombus ABCD, $\overline{AB} = 5\text{ cm}$, $m\angle A = 60^\circ$.

Required: To construct Rhombus ABCD from the given data.

Steps of construction

1. Draw AB, 5 cm long.
2. At point A, construct $\angle BAX$ such that $m\angle BAX = 60^\circ$.
3. Taking point A as centre, with radius $\overline{AB}$, draw an arc to intersect AX at point D.
4. Taking points B and D as centres, with the same radius of $\overline{AB}$, draw arcs cutting each other at C.
5. Draw $\overline{BC}$ and $\overline{DC}$.

Now Quad ABCD is the required Rhombus.



When one side and a diagonal are given

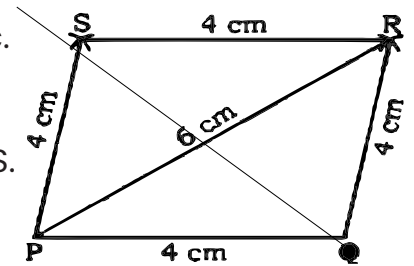
Example 2. Construct Rhombus PQRS when $\overline{PQ} = 4\text{ cm}$ and $\overline{PR} = 6\text{ cm}$.

Given: In Rhombus PQRS, $\overline{PQ} = 4\text{ cm}$, $\overline{PR} = 6\text{ cm}$.

Required: To construct Rhombus PQRS from the given data.

Steps of construction:

1. Draw PQ, 4 cm long.
2. Take point P as centre, with radius 6 cm, draw an arc.
3. Take point Q as centre and draw an arc of radius 4 cm that cuts the previous arc at R.
4. Repeat the above step by taking P as centre to find S.
5. Join point P to R and Q to R. (Draw $\overline{PR}$ and $\overline{QR}$)
6. Join point S to P and R to S. (Draw $\overline{PS}$ and $\overline{SR}$)
7. Now Quadrilateral PQRS is the required Rhombus.



CONSTRUCT A PARALLELOGRAM

(a) When two diagonals and angles between them are given.

(b) When two adjacent sides and the angle included between them is given.

When two diagonals and angles between them are given.

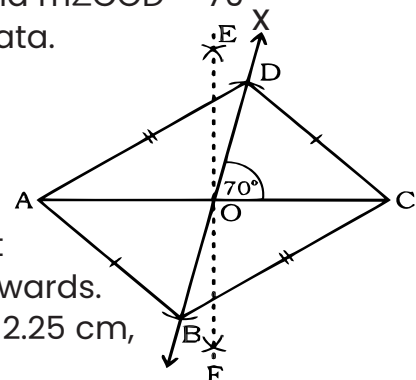
Example 1. Construct a $\parallel^m$ ABCD, in which $\overline{AC} = 6\text{ cm}$, $\overline{BD} = 4.5\text{ cm}$ and angle between them is of 70° .

Given: In $\parallel^m$ ABCD, $\overline{AC} = 6\text{ cm}$, $\overline{BD} = 4.5\text{ cm}$ and $m\angle COD = 70^\circ$

Required: To construct $\parallel^m$ ABCD from the given data.

Steps of construction:

1. Draw $\overline{AC}$, 6 cm long.
2. Draw $\overline{EF}$ the right bisector of $\overline{AC}$ intersecting AC at point O.
3. At point O, with the help of protractor construct $\angle COX$ of measure 70° . Also produce the line backwards.
4. Take point O as centre and with radius $4.5/2 = 2.25\text{ cm}$, draw arcs on both sides to cut $\overline{OX}$ at D and B.



5. Draw $\overline{AB}$, $\overline{AD}$, $\overline{BC}$ and $\overline{CD}$.

Now, Quad ABCD is the required parallelogram.

When two adjacent sides and the angle included between them is given.

Example 2. Construct $\parallel^m$ WXYZ in which $m\overline{WX} = 6$ cm, $m\overline{WZ} = 4$ cm and $m\angle XWZ = 120^\circ$.

Given: In $\parallel^m$ WXYZ, $m\overline{WX} = 6$ cm, $m\overline{WZ} = 4$ cm and $m\angle XWZ = 120^\circ$

Required: To construct $\parallel^m$ WXYZ from the given data.

Steps of construction:

1. Draw $\overline{WX}$, 6 cm long.

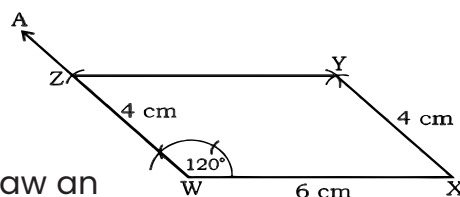
2. At point W, construct $\angle XWA$ such that $m\angle XWA = 120^\circ$.

3. Take point W as centre, with radius 4 cm, draw an arc to cut WA at Z.

4. Take point X as centre and with radius 4 cm, draw an arc.

5. Take point Z as centre and with radius 6 cm, draw another arc to cut the previous arc at Y.

Now, Quad WXYZ is the required parallelogram.



CONSTRUCT A KITE

When two unequal sides and a diagonal are given:

Example 3. Construct Kite ABCD in which $m\overline{AB} = 3$ cm, $m\overline{BC} = 5$ cm and diagonal $\overline{BD} = 4$ cm

Given: In Kite ABCD, $m\overline{AB} = 3$ cm, $m\overline{BC} = 5$ cm and $m\overline{BD} = 4$ cm

Required: To construct Kite ABCD from the given data.

Steps of construction:

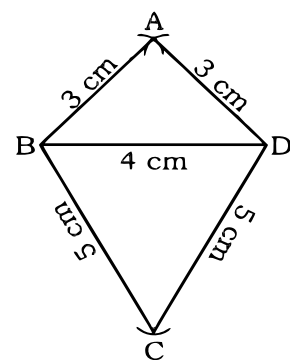
1. Draw BD, 4 cm long.

2. Take point B and D as centres and with radius 3 cm, draw arcs to cut one another at point A.

3. Again take points B and D as centres and with radius 5 cm, draw arcs to cut one another at point C.

4. Draw $\overline{AB}$, $\overline{AD}$, $\overline{BC}$ and $\overline{DC}$.

Now Quad ABCD is the required Kite.



CONSTRUCT A REGULAR PENTAGON

When measurement of side is given

Example 4. Construct a Regular Pentagon ABCDE, each of whose side measures 4 cm.

Given: In Regular Pentagon ABCDE, $m\overline{AB} = 4$ cm

$$m\angle A = m\angle B = \left(\frac{2(5) - 4}{5}\right) 90^\circ$$

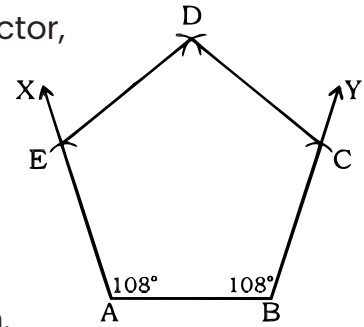
$$= \left(\frac{10 - 4}{5} \right) \times 90^\circ = \frac{6}{5} \times 90^\circ = 108^\circ$$

Required: To construct Regular Pentagon ABCDE from the given data.

Steps of construction:

1. Draw AB, 4 cm long.
2. At points A and B, construct, with the help of protractor, $\angle BAX$ and $\angle ABY$ such that each angle measures 108° .
3. Take points A and B as centres, and with radius mAB , draw arcs to cut $\overrightarrow{AX}$ at E and $\overrightarrow{BY}$ at point C.
4. Take points C and E as centres and with the same radius, draw arcs to cut one another at point D.
5. Draw CD and ED.

Now polygon ABCDE is the required Regular Pentagon.



CONSTRUCT A REGULAR HEXAGON

When a side is given

Example 5. Construct Regular Hexagon ABCDEF each of whose sides measures 3 cm.

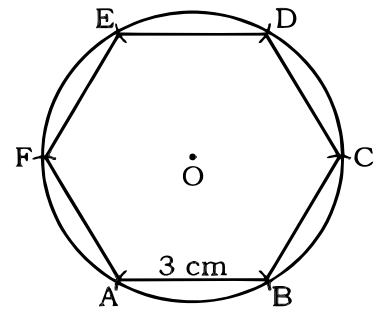
Given: In Regular Hexagon, $mAB = 3$ cm

Required: To construct Regular Hexagon ABCDEF from the given data.

Steps of construction:

1. At a convenient place, take point O as centre and with radius 3 cm construct a circle.
2. Take a point A on the circle.
3. Take point A as centre, with radius 3 cm, draw an arc to cut the circle at B.
4. Then taking point B as centre, with the same radius, draw another arc to cut the circle at C.
5. Again take point C as centre and with the same radius, draw an arc to cut the circle at point D and repeat the same process with Centres D and E to cut the circle at points E and F respectively.
6. Draw AB, BC, CD, DE, EF, and FA.

Now polygon ABCDEF is the required Regular Hexagon.



EXERCISE 8.1

1. Draw similar converging lines in your exercise book as given below, then:

- (i) Draw the bisector of the angle formed by the converging lines
- (ii) Find the measure of the angle without producing the lines.

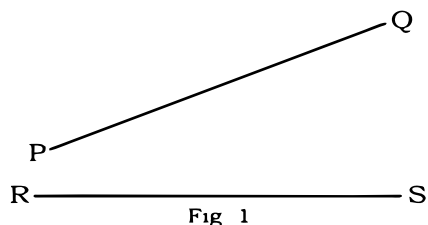


Fig. 1

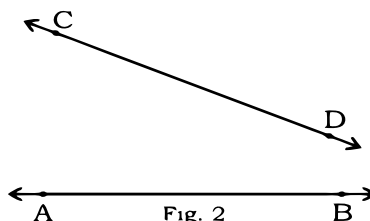


Fig. 2

2. (a) Construct square ABCD in which the measure of its diagonal is as under:

- (i) $m\overline{AC} = 6$ cm
- (ii) $m\overline{BD} = 7$ cm
- (iii) $m\overline{AC} = 8$ cm

(b) Construct square PQRS when the difference between its diagonal and a side is:

- (i) 2.5 cm
- (ii) 3 cm
- (iii) 3.5 cm
- (iv) 4.5 cm

(c) Construct square WXYZ when the sum of its diagonal and a side is:

- (i) 10 cm
- (ii) 12 cm
- (iii) 11 cm
- (iv) 8 cm

3. (a) Construct rectangle PQRS when its two sides are:

- (i) $m\overline{PQ} = 6$ cm, $m\overline{QR} = 4$ cm
- (ii) $m\overline{PQ} = 4$ cm, $m\overline{QR} = 7$ cm

(b) Construct rectangle ABCD, when its diagonal and one side are given:

- (i) $m\overline{AC} = 6$ cm, $m\overline{AB} = 4$ cm
- (ii) $m\overline{BD} = 7$ cm, $m\overline{AB} = 5$ cm

4. (a) Construct Rhombus ABCD when one side and base angle are given:

- (i) $m\overline{AB} = 4$ cm, $m\angle A = 120^\circ$
- (ii) $m\overline{AB} = 5$ cm, $m\angle A = 60^\circ$

(b) Construct Rhombus PQRS, when one side and a diagonal are given:

- (i) $m\overline{PQ} = 5$ cm, $m\overline{QS} = 4$ cm
- (ii) $m\overline{PQ} = 4.5$ cm, $m\overline{PR} = 7$ cm

5. (a) Construct $\parallel$ PQRS, when two diagonals and angle between them are given:

- (i) $m\overline{QR} = 8$ cm, $m\overline{QS} = 6$ cm, $m\angle QOR = 60^\circ$
- (ii) $m\overline{PR} = 6$ cm, $m\overline{QS} = 8$ cm, $m\angle QOR = 120^\circ$

(b) Construct $\parallel$ ABCD, when two adjacent sides and included angle are given:

- (i) $m\overline{AB} = 6$ cm, $m\overline{BC} = 4$ cm, $m\angle B = 70^\circ$

(ii) $m\overline{AB} = 4$ cm, $m\overline{AD} = 5$ cm, $m\angle A = 100^\circ$

6. Construct Kite ABCD, when two unequal sides and a diagonal are given:

(i) $m\overline{AB} = 4$ cm, $m\overline{BC} = 7$ cm, $m\overline{AC} = 9$ cm

(ii) $m\overline{AB} = 4$ cm, $m\overline{BC} = 6$ cm, $m\overline{BD} = 4$ cm

WORKSHEET

1. Construct a parallelogram ABCD in which $AB = 5.2$ cm, $BC = 4.7$ cm and $AC = 7.6$ cm.
2. Construct a parallelogram ABCD in which $AB = 4.3$ cm, $AD = 4$ cm and $BD = 6.8$ cm.
3. Construct a parallelogram PQRS in which $QR = 6$ cm, $PQ = 4$ cm and $\angle PQR = 60^\circ$.
4. Construct a parallelogram ABCD in which $BC = 5$ cm, $\angle BCD = 120^\circ$ and $CD = 4.8$ cm.
5. Construct a parallelogram, one of whose sides is 4.4 cm and whose diagonals are 5.6 cm and 7 cm. Measure the other side.
6. Construct a rhombus ABCD in which $AB = 4$ cm and diagonal AC is 6.5 cm.
7. Draw a rhombus whose side is 7.2 cm and one angle is 60° .

CONSTRUCTION OF RIGHT ANGLE TRIANGLE

Construct a right angled triangle when hypotenuse and one side are given:

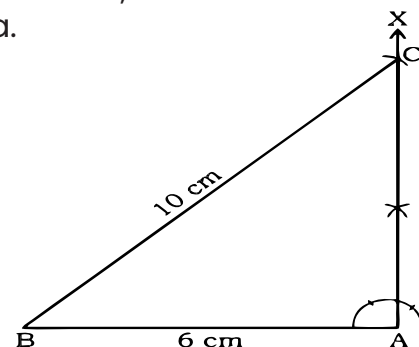
Example 1. Construct right-angled Triangle ABC, in which hypotenuse $m\overline{BC} = 10$ cm and $m\overline{AB} = 6$ cm

Given: In right-angled $\triangle ABC$, $m\overline{BC} = 10$ cm, $m\overline{AB} = 6$ cm, $m\angle A = 90^\circ$

Required: To construct $\triangle ABC$ from the given data.

Steps of construction:

1. Draw AB , 6 cm long.
 2. At point A , draw $AX \perp AB$.
 3. Take point B as centre and with radius 10 cm, draw an arc which cuts AX at point C .
 4. Join points B and C . (Draw BC)
- Now $\triangle ABC$ is the required right triangle.



(a) Construct a right angled triangle when the hypotenuse and the Vertical Height from the vertex to the Hypotenuse are given:

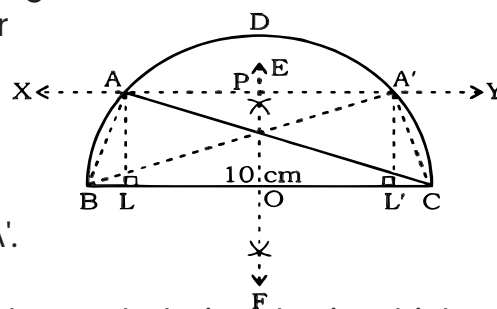
Example 2. Construct a right-angled $\triangle ABC$, in which hypotenuse $\overline{BC}$ is 10 cm and vertical height from point A to BC is 3.8 cm.

Given: In right-angled $\triangle ABC$, $m\angle A = 90^\circ$, $mBC = 10$ cm, $mAL = 3.8$ cm ($AL \perp BC$)

Required: To construct $\triangle ABC$ from the given data.

Steps of construction:

1. Draw BC , 10 cm long.
2. Draw EF , the right bisector of BC intersecting it at O .
3. Take O as centre, and with radius mOC or mOB ($10/2=5$ cm), draw a semi-circle BDC .
4. Take a point P on OE such that $mOP = 3.8$ cm
5. Through point P , draw $XY \parallel BC$ (i.e., draw $XY \perp OP$) cutting the circle at points A and A' .
6. Draw AB and AC (Also draw $A'B$ and $A'C$)



Now $\triangle ABC$ (and $\triangle A'BC$) are the required right-angled triangles in which $mAL = mAL' = 3.8$ cm

Note: In any semi-circle, inscribed triangle is a right-angled triangle.

EXERCISE 8.2

1. Construct the following right-angled triangles when:

- (a) Hypotenuse = 10 cm and measure of one side = 8 cm
- (b) Hypotenuse = 13 cm and measure of one side = 12 cm
- (c) One side $PQ = 5$ cm, one side $SQ = 6$ cm

2. Construct the following right-angled triangles when:

- (a) Hypotenuse = 10 cm and vertical height from vertex to Hypotenuse = 3 cm
- (b) Hypotenuse = 12 cm and vertical height from vertex to Hypotenuse = 4 cm

3. Construct the right-angled triangles ABC and DBC on either sides of Hypotenuse BC, 10 cm long when:

- (a) Vertical height $AL = 3.5$ cm, $mDC = 5$ cm
- (b) Vertical height $AL = 3$ cm, $mDB = 6$ cm

Construction of Right Angles Triangle when its Hypotenuse and one side are given

1. Construct a $\triangle ABC$ in which $AC = CB = 7.3$ cm, and $\angle B = 45^\circ$. Is this a right-angle triangle?
2. Draw a right-angle triangle $\triangle ABC$, right-angled at A , with $AB = 3$ cm and $BC = 5$ cm.

3. Construct a right-angled triangle whose hypotenuse is 6 cm long and one of the legs is 4 cm long.
4. Construct a triangle $\triangle WAS$ with $\angle W = 90^\circ$, $WA = 6$ cm and $\angle S = 50^\circ$.
5. Construct a triangle $\triangle ABC$ with $\angle B = 90^\circ$, $AB = 8$ cm and $BC = 6$ cm.
6. Construct a triangle $\triangle ISR$ with $\angle I = 90^\circ$, $IS = 4$ cm and $SR = 8$ cm.

Q. Using a ruler and protractor/angle ruler, draw and label the triangle with the following properties.

1. $AB = 2$ in., $\angle CAB = 45^\circ$ and $AC = 1.5$ in.

2. $\angle ABC = 45^\circ$, $\angle BCA = 45^\circ$, and $BC = 2.5$ in.

3. $\angle BCA = 60^\circ$, $BC = 1.5$ in., and $AC = 1.5$ in.

4. $\angle ABC = 50^\circ$, $\angle BCA = 70^\circ$, and $BC = 2$ in.